Reinforcement Learning - Sutton and Barto

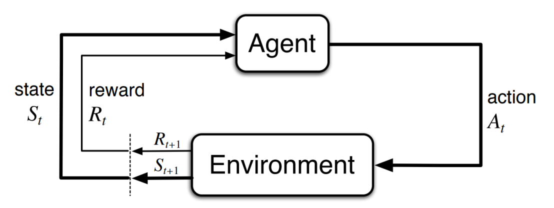
Book Club Highlights

Alexandra Ortan

December 11, 2025

Chapter 3
Finite Markov Decision Processes

Agent-Environment Interface



agent learner and decision maker

environment everything outside the agent

$S_t \in \mathcal{S}$ state of the environment at time t

$A_t \in \mathcal{A}(s)$ action at time t

$R_t \in \mathcal{R} \subset \mathbb{R}$ reward received as a consequence of action A_{t-1}

Markov Decision Processes (MDP)

Markov Property

The probability of each possible value for S_t and R_t is completely determined by the preceding state and action S_{t-1} and A_{t-1}.

Dynamics Function

The function $p : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$ is defined as

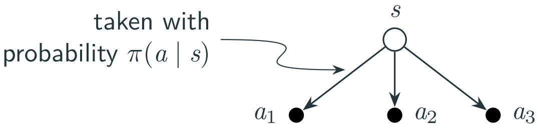
$$p(s', r \mid s, a) \doteq Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

and completely defines the dynamics of the MDP.

Policy

Mapping from states to probability of selecting each possible action.

$$\pi : \mathcal{S} \mapsto \{\text{probabilities over } \mathcal{A}(s)\}$$



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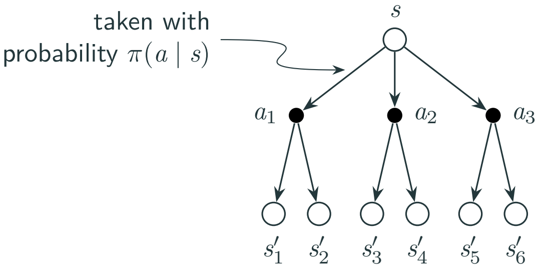

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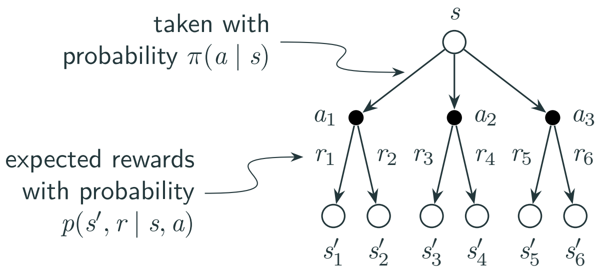
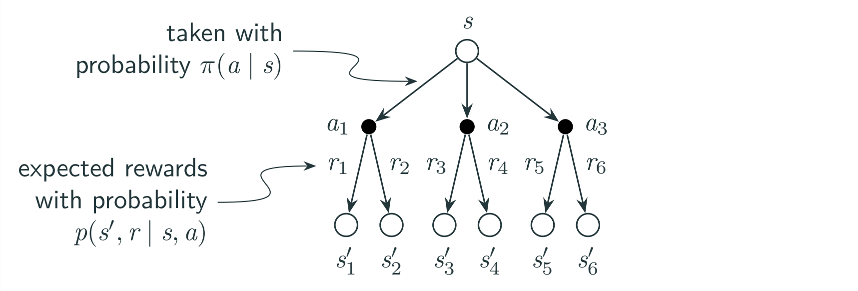


State value function

Expected return when starting in a given state s and following policy π thereafter.

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$



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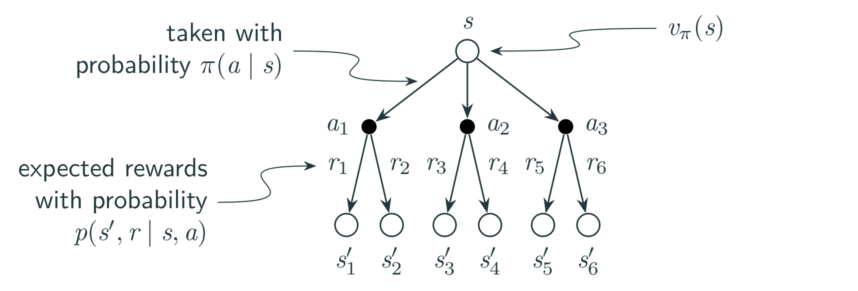
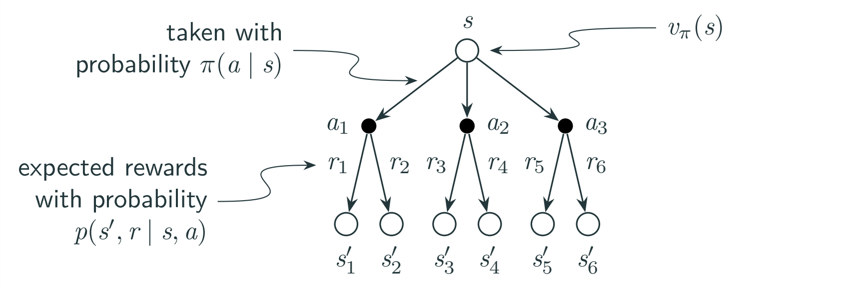


Action value function

Expected return when starting in a given state s, taking an action a, then following policy π thereafter.

$$q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$



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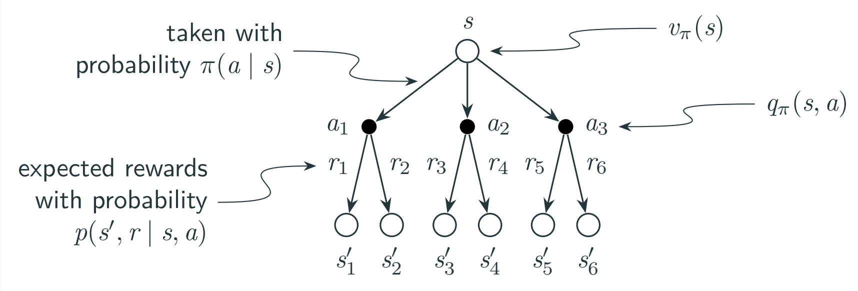
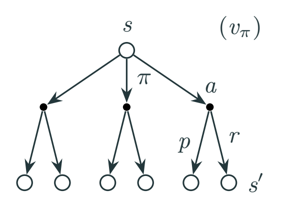


Bellman Equations

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_\pi(s') \right]$$



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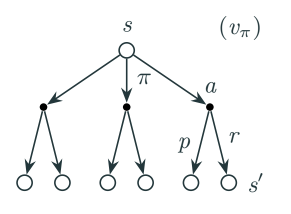

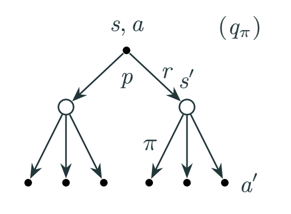

Policies and Value Functions

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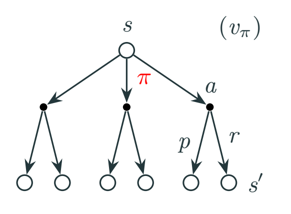

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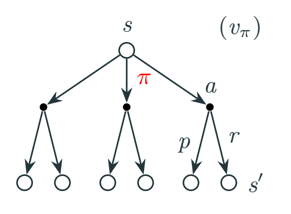

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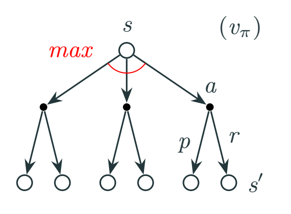

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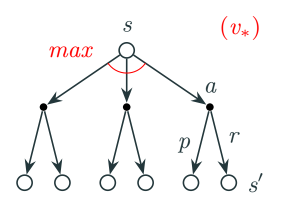

Optimal Policies and Optimal Value Functions

Optimal Policy

π_* is said to be optimal if its expected return is greater than or equal to that of all other policies for all states:

$$v_{\pi_*}(s) \geq v_{\pi'}(s) \qquad \forall s \in \mathcal{S}, \forall \text{ policies } \pi'$$

Optimal Value Functions

Value functions associated with optimal policies:

$$\begin{aligned} v_*(s) &= v_{\pi_*}(s) &&= \max_\pi v_\pi(s) &&\forall s \in \mathcal{S} \\ q_*(s,a) &= q_{\pi_*}(s,a) &&= \max_\pi q_\pi(s,a) &&\forall a \in \mathcal{A}(s), \forall s \in \mathcal{S} \end{aligned}$$

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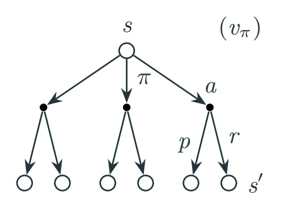

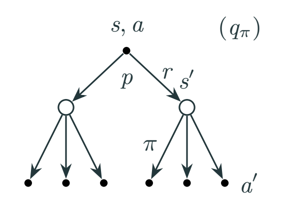

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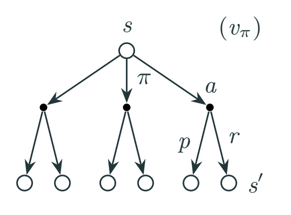

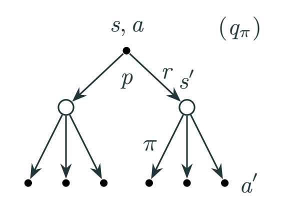

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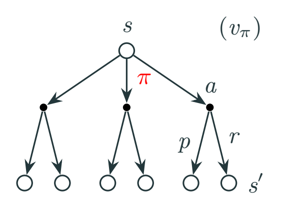

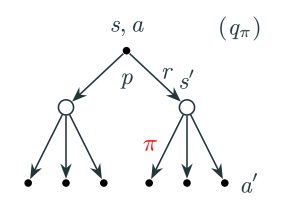

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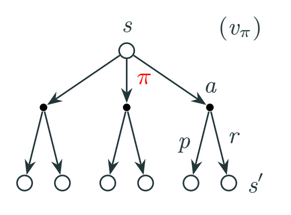

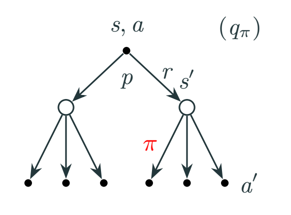

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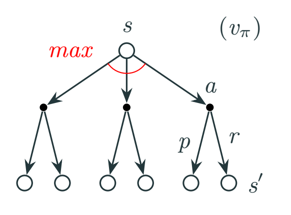

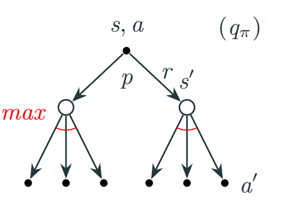

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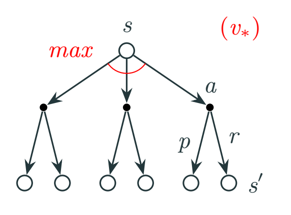

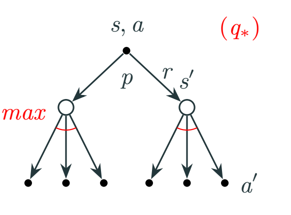

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Bellman Optimality Equations

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